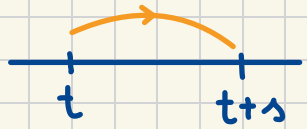
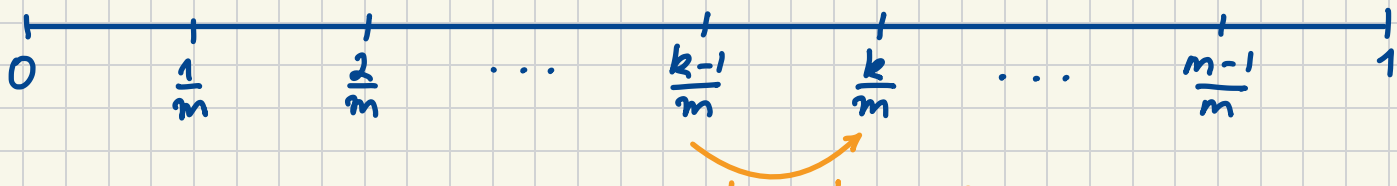
M339D: March 6th, 2026.

More on Realized Returns.

Review:



$$\boxed{R(t, t+s) := \ln\left(\frac{S(t+s)}{S(t)}\right)}$$

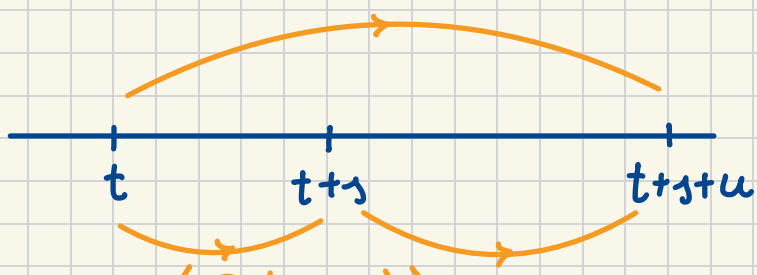


$R\left(\frac{k-1}{m}, \frac{k}{m}\right)$ for $k = 1..m$

are random variables

We make the following assumptions:

- the returns over disjoint intervals are independent;
- all the returns over the same length intervals are identically distributed.



$$R(t, t+s+u) = \ln\left(\frac{S(t+s+u)}{S(t)}\right)$$

$$= \ln\left(\frac{S(t+s)}{S(t)} \cdot \frac{S(t+s+u)}{S(t+s)}\right)$$

$$= \boxed{\ln\left(\frac{S(t+s)}{S(t)}\right)} + \boxed{\ln\left(\frac{S(t+s+u)}{S(t+s)}\right)}$$

$$= R(t, t+s) + R(t+s, t+s+u)$$

Hence, realized returns are additive

add.

$$\Rightarrow \quad R(0,\tfrac{1}{m}) + R(\tfrac{1}{m},\tfrac{2}{m}) + \cdots + R(\tfrac{m-1}{m},1) = R(0,1)$$

Q: $Var[R(0,1)] = \underline{\sigma^2}$

$$\Rightarrow \quad \sigma^2 = Var[R(0,\tfrac{1}{m}) + \cdots + R(\tfrac{m-1}{m},1)] \quad \text{(independence)}$$

$$= Var[R(0,\tfrac{1}{m})] + \cdots + Var[R(\tfrac{m-1}{m},1)] \quad \text{(identically dist'd)}$$

$$= m \cdot Var[R(0,\tfrac{1}{m})] = m \cdot \sigma_{h_m}^2$$

$$\Rightarrow \quad \sigma_{h_m}^2 = \frac{1}{m}\sigma^2$$

$$\Rightarrow \quad \boxed{\sigma_{h_m} = \sigma\sqrt{\frac{1}{m}} = \sigma\sqrt{h_m}}$$

We generalize this identity to arbitrary lengths h:

$$\boxed{\sigma_h = \sigma\sqrt{h}}$$

Forward Binomial Trees [revisited].

$$\underbrace{S(0) \to S_u = u \cdot S(0),\ F_{0,h}(S),\ S_d = d \cdot S(0)}_{h}$$

No arbitrage condition:

$$d < e^{rh} < u$$

Recall: $F_{0,h}(S) = S(0)e^{rh}$

$$S_u := F_{0,h}(S)e^{\sigma\sqrt{h}} = S(0)e^{rh} \cdot e^{\sigma\sqrt{h}} = S(0)\underbrace{\boxed{e^{rh+\sigma\sqrt{h}}}}_{u}$$

$$S_d := F_{0,h}(S)e^{-\sigma\sqrt{h}} = S(0)e^{rh} \cdot e^{-\sigma\sqrt{h}} = S(0)\underbrace{\boxed{e^{rh-\sigma\sqrt{h}}}}_{d}$$

Q: Do u and d satisfy the no-arbitrage condition?

$\longrightarrow$: $d < e^{rh} < u$

$$e^{rh-\sigma\sqrt{h}} < e^{rh} < e^{rh+\sigma\sqrt{h}}$$

$$\boxed{e^{-\sigma\sqrt{h}} < 1 < e^{\sigma\sqrt{h}}} \qquad \sigma, h > 0 \quad \square$$

Q: What is $\frac{S_u}{S_d} = ?$

$\longrightarrow$: $$\frac{S_u}{S_d} = \frac{\cancel{S(0)}\cancel{e^{rh}} \cdot e^{\sigma\sqrt{h}}}{\cancel{S(0)}\cancel{e^{rh}} \cdot e^{-\sigma\sqrt{h}}} = e^{2\sigma\sqrt{h}}$$

<u>Example</u>. Consider a one-period <u>forward</u> binomial tree w/ period length of <u>one quarter</u>.

$$100 = S(0) \begin{cases} S_u = \underline{105} \\ S_d = \underline{98} \end{cases}$$

Q: What is the volatility of this stock?

$\longrightarrow$: $$\frac{S_u}{S_d} = \frac{105}{98} = e^{\cancel{2}\sigma\sqrt{\cancel{\frac{1}{4}}}}$$

$$\boxed{\sigma = \ln\left(\frac{105}{98}\right)}$$

$\square$

<u>Binomial Option Pricing.</u>

<u>Payoff</u>

$$\underbrace{\begin{array}{l} S(0) \\ V(0) = ? \end{array} \begin{cases} S_u = u \cdot S(0) \\ S_d = d \cdot S(0) \end{cases}}_{h}$$

$V_u = v(S_u)$

$V_d = v(S_d)$

populating the tree $\longrightarrow$

$\longleftarrow$ PRICING

<u>Goal</u>: Pricing a European style derivative security w/ exercise date @ the end of tree, i.e., $\boxed{T = h}$

It's completely determined by its <u>payoff f'tion: $v(\cdot)$</u>

<u>e.g.</u>, for a call: $v_c(s) = (s-K)_+$

for a put: $v_c(s) = (K-s)_+$

for a power option: $v(s) = (s^2-K)_+$

The payoff of our derivative security is a random variable

$$\boxed{V(T) := v(S(T))}$$